

Bajaj Auto OHM Campus challenge 2026

Stage II | Team “Chem_Chrew”

Project PRANA:

Next-Generation Commercial
3-Wheeler EV Strategy

IIT
Kharagpur



**ABHINAM
DUTTA**

B.Tech 2nd year

IIT
Kharagpur



**UJJAL
BARMAN**

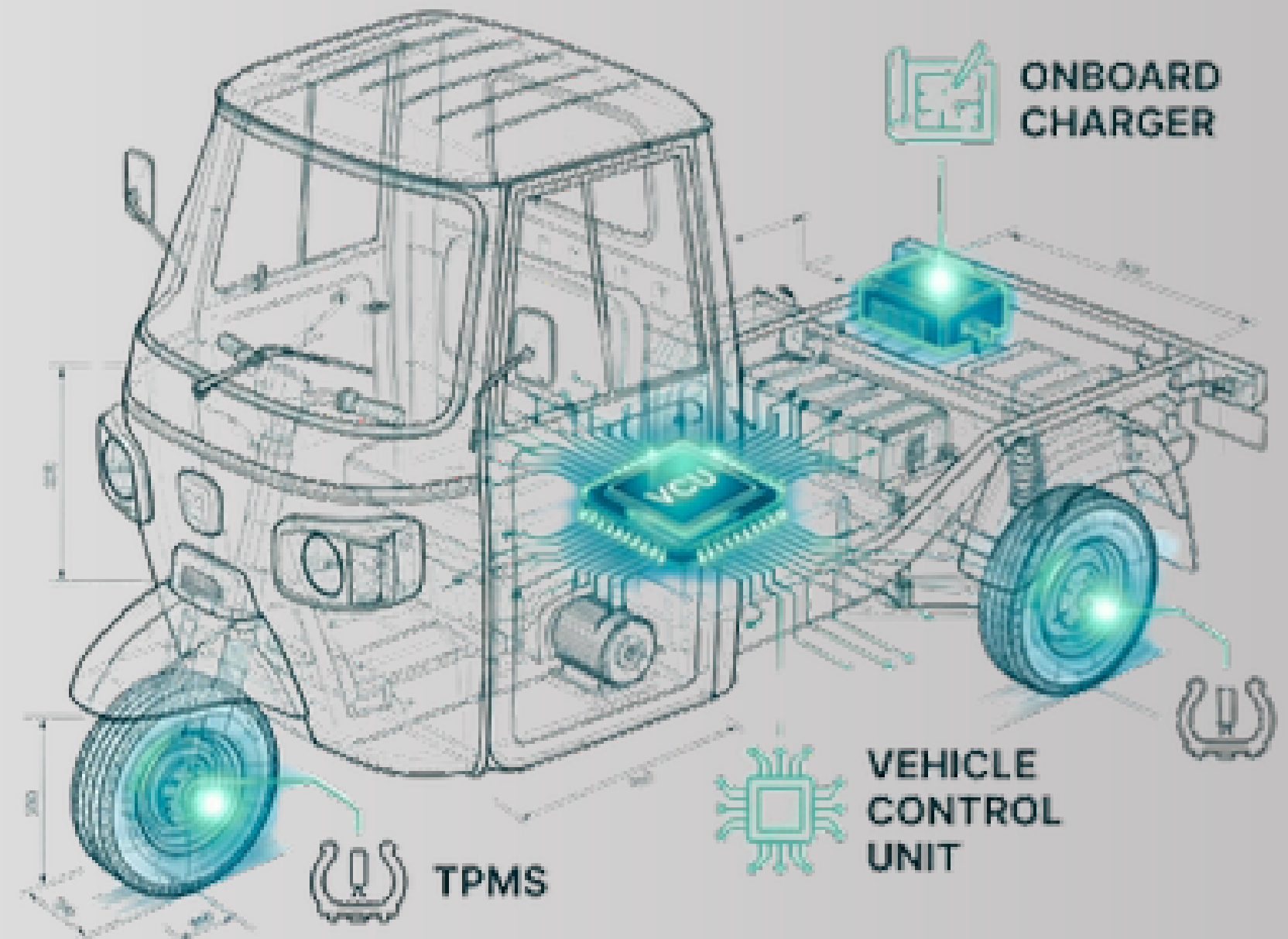
B.Tech 2nd year

IIT
Kharagpur



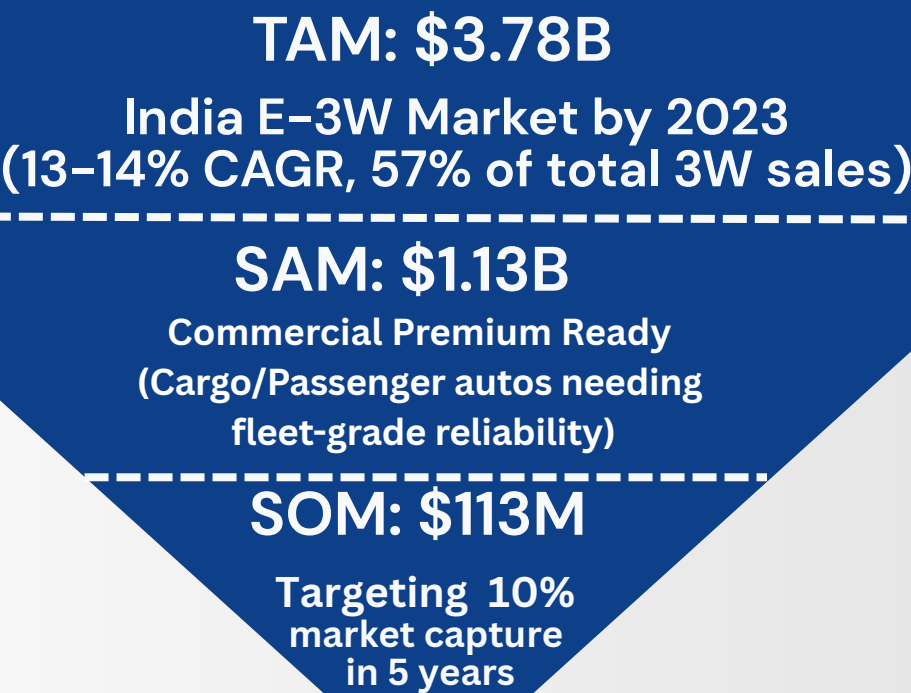
**ROUNAK
CHHATAIT**

B.Tech 2nd year



A \$3.78B Market. A 24-Month Window.

TAM/SAM/SOM Market Sizing



Macro-Environmental PESTLE Framework

Political

FAME III / PM
E-Drive subsidies
linking to telematics
compliance.

Economic

Extreme TCO
sensitivity; daily
usage determines
livelihood.

Social

Operator income
strictly tied to
uptime (12-hr shifts,
6 days/wk).

Technology

LFP battery maturity
enables native 0.44C
moderate fast-
charging.

Legal

Mandatory TPMS
scaling from 4W (AIS
159) to 3W in 24-36
months.

Environmental

Commercial EVs
displacing ICE;
potential 23,046
tonnes CO2 offset.

Operator Persona Economics

Persona A

(Shivaraj/owner)
110-130 km/day
₹42K monthly
gross 3.8K EMI

Unpredictable
downtime wipes
out margins.

Persona B

(Ranjit/Small Fleet)
4 vehicles
₹1.6-2.0L/mo revenue
Depot charging

Lacks fleet health
visibility.

Persona C

(Aarav/Large Fleet)
18 vehicles
₹9.5-12L/mo
revenue

Requires predictive
maintenance &
compliance data.

3C's Strategic Fit Framework

Company

Bajaj's unmatched
scale and R&D native
hardware/software
integration.

**Conclusion: 24-36
Month First-Mover
Window is Critical**

Competitor

Legacy players bolting on
disjointed features; lacking
unified architecture.

Customer

Commercial operators
demand pure
economic ROI
(uptime>features)

Problem & Root Cause

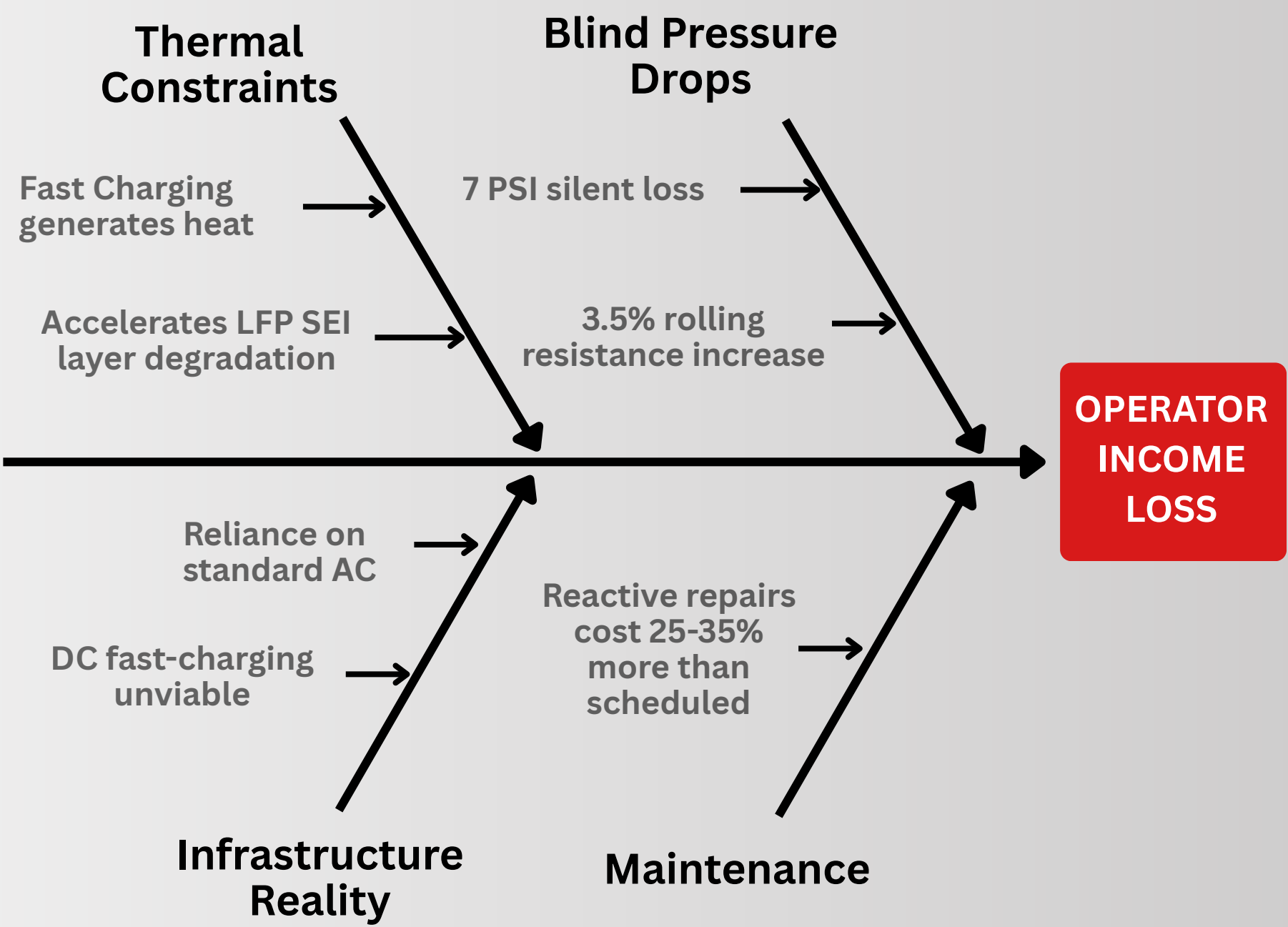
Operator Persona Economics

Persona A
(Shivraj/Owner)
₹42K monthly gross
₹3.8K EMI
Unpredictable downtime
wipes out margins.

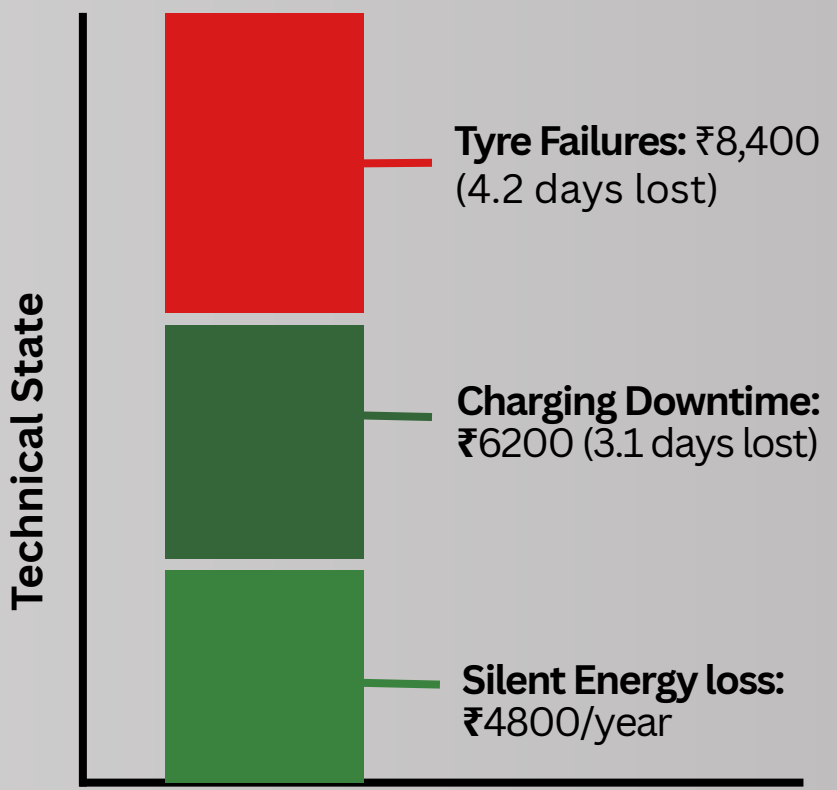
Persona B
(Shivraj/Owner)
4 vehicles
₹1.6-2.0L/mo revenue
Depot Charging
Lacks fleet health visibility

Persona C
(Aarav/Large Fleet)
18 vehicles
₹9.5-12L/mo revenue
Requires predictive
maintenance & compliance
data

Root-Cause Analysis



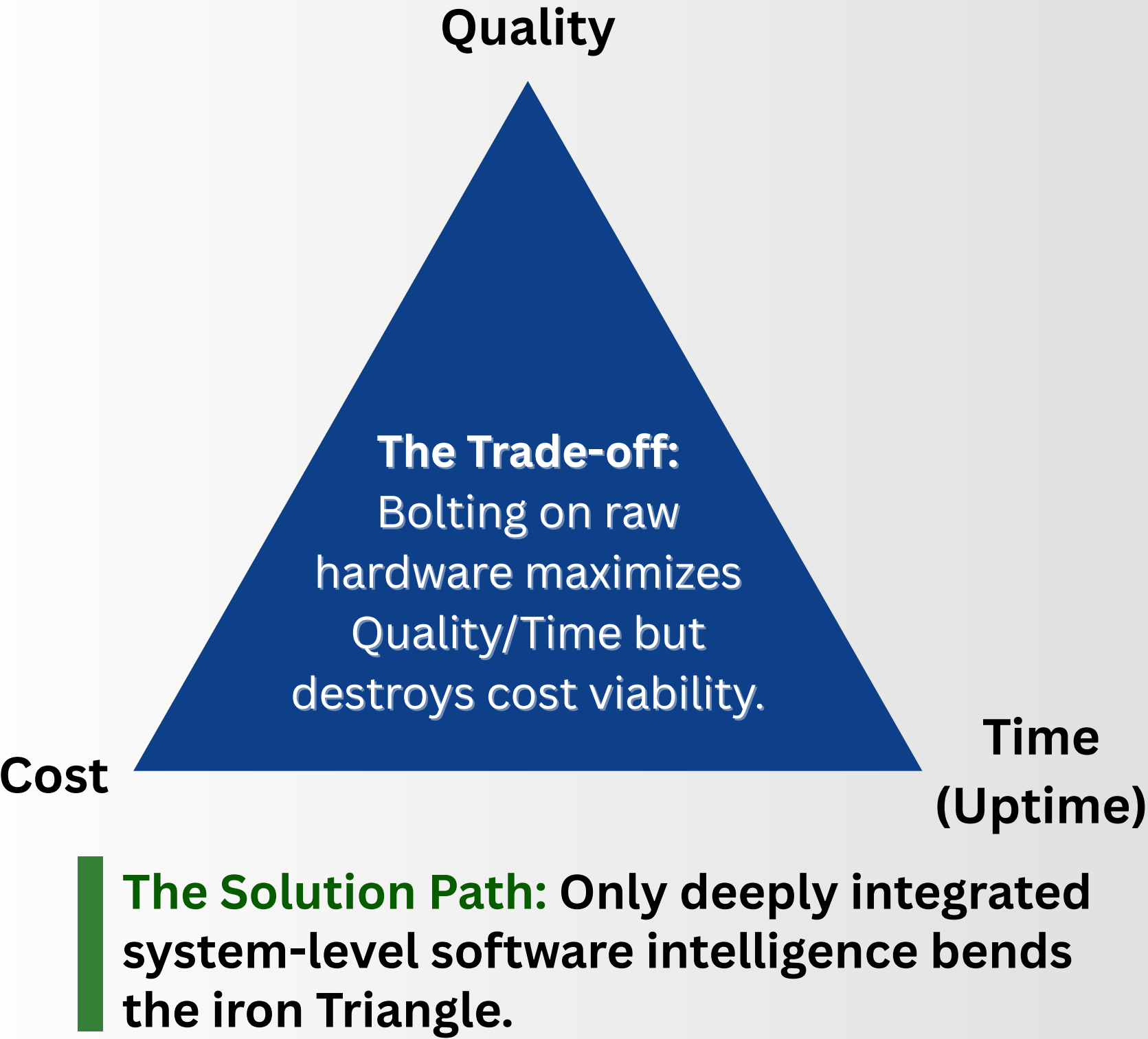
The Cost of Doing Nothing



Total Deficit:
₹27,700/year and 11.6
days of lost income.

Architecture & Recommendation

Iron Triangle / Triple Constraint

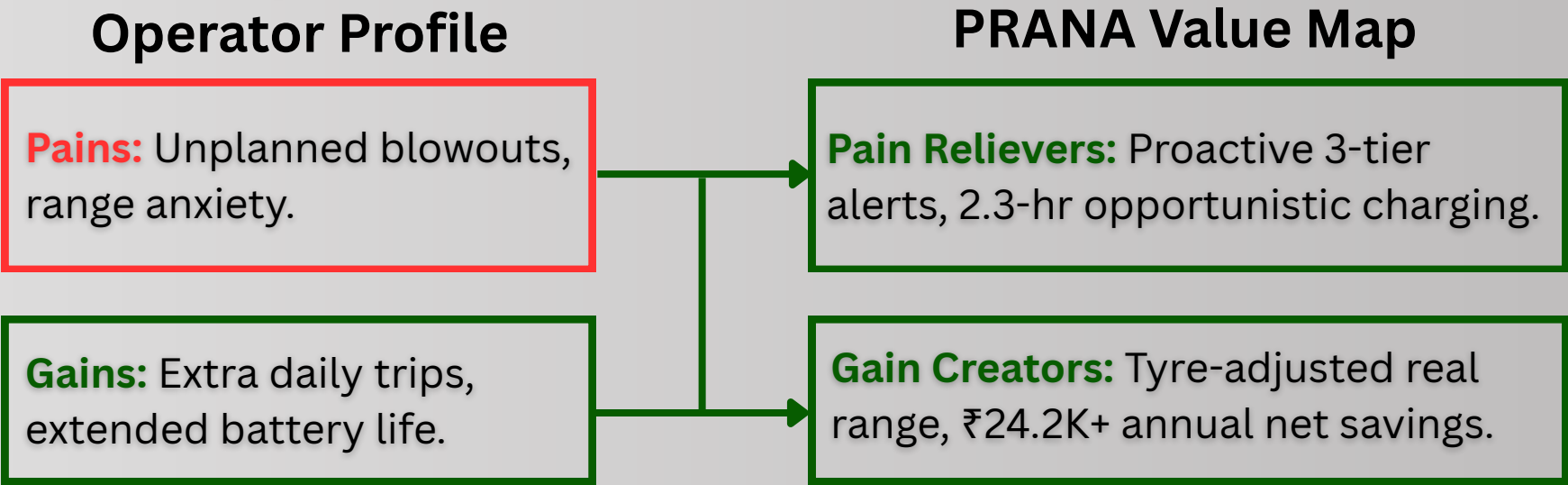


Pugh Decision Matrix

Criteria	Weight	Concept (Base)	Concept (Mod)	Concept 3 (Prem)	PRANA (Rec)
Cost	25%	5	4	1	4
Feasibility	20%	5	2 Note: No ABS	3	5
Scalability	15%	4	3	2	5
Safety	25%	3	3	5	5
Performance	15%	2	4	5	5

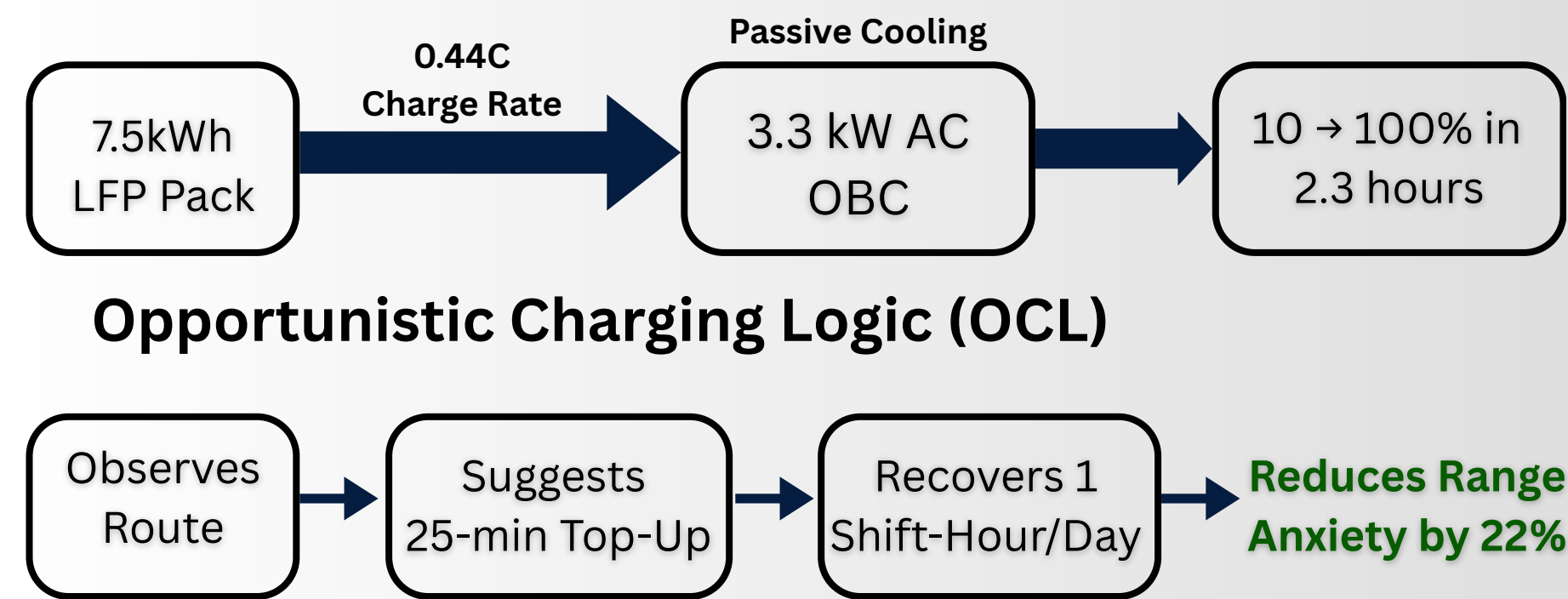
Recommendation: Modified Concept 2 merging 3.3kW AC Fast Charging with Direct BLE TPMS via an Edge-Compute Intelligence Layer.

Value Proposition Canvas

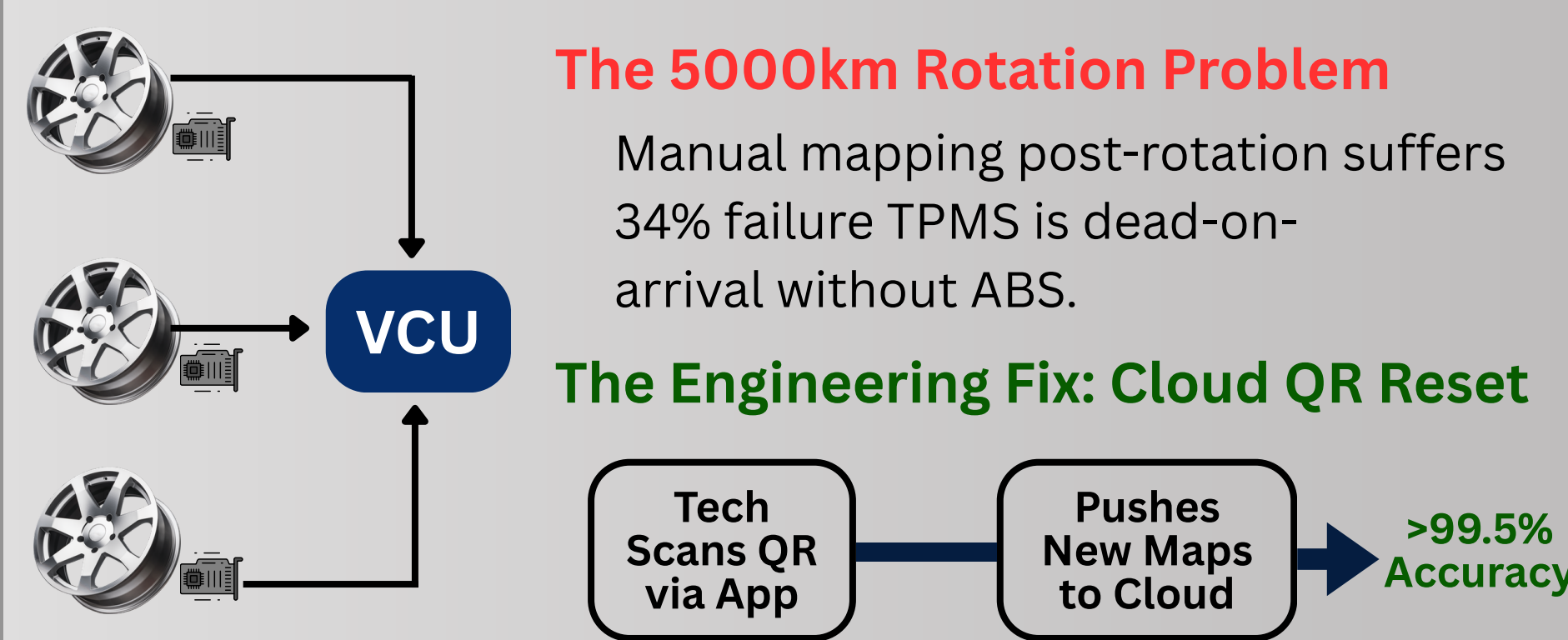


Core Sub-Systems Design

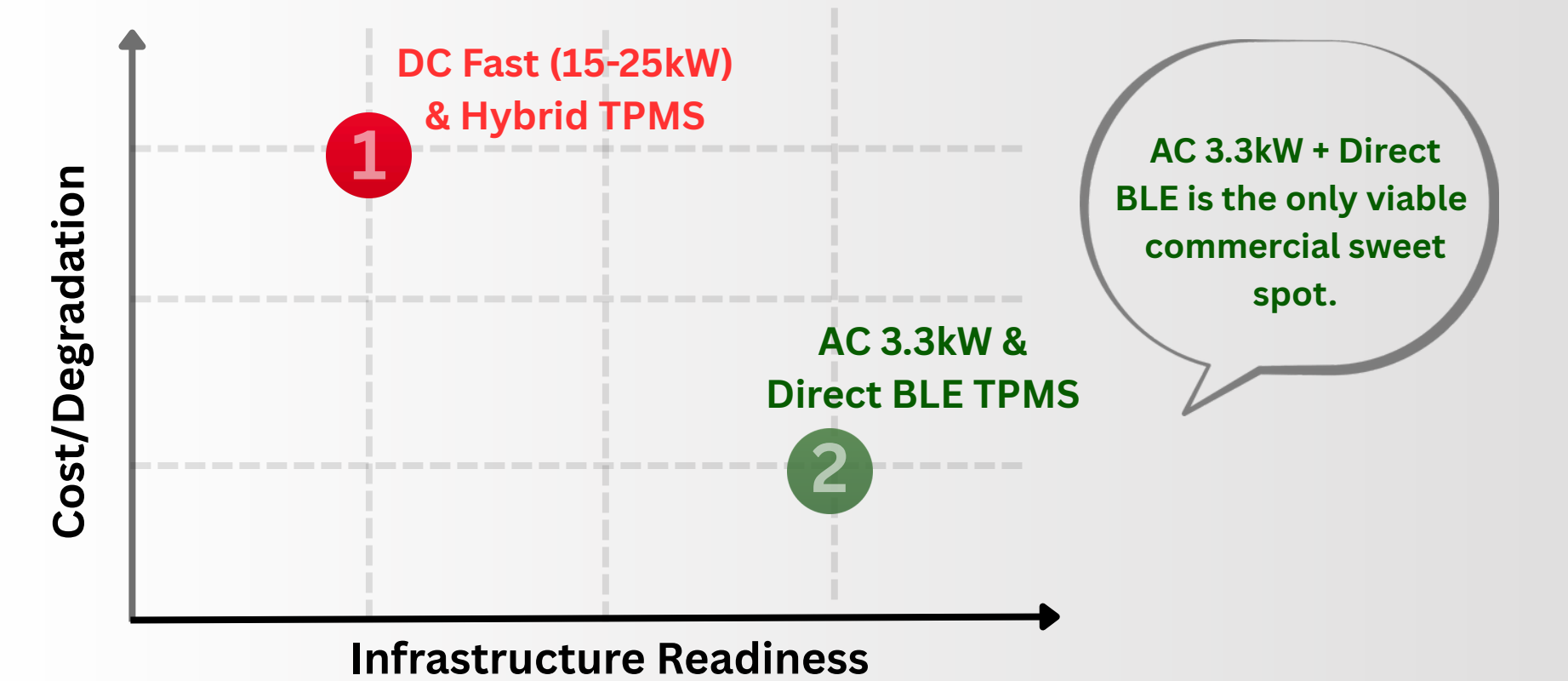
Core-System 01: Smart Charging Core (SCC)



Sub-System 02: Intelligent Tyre System (ITS)



Technical Charging Logic (OCL)

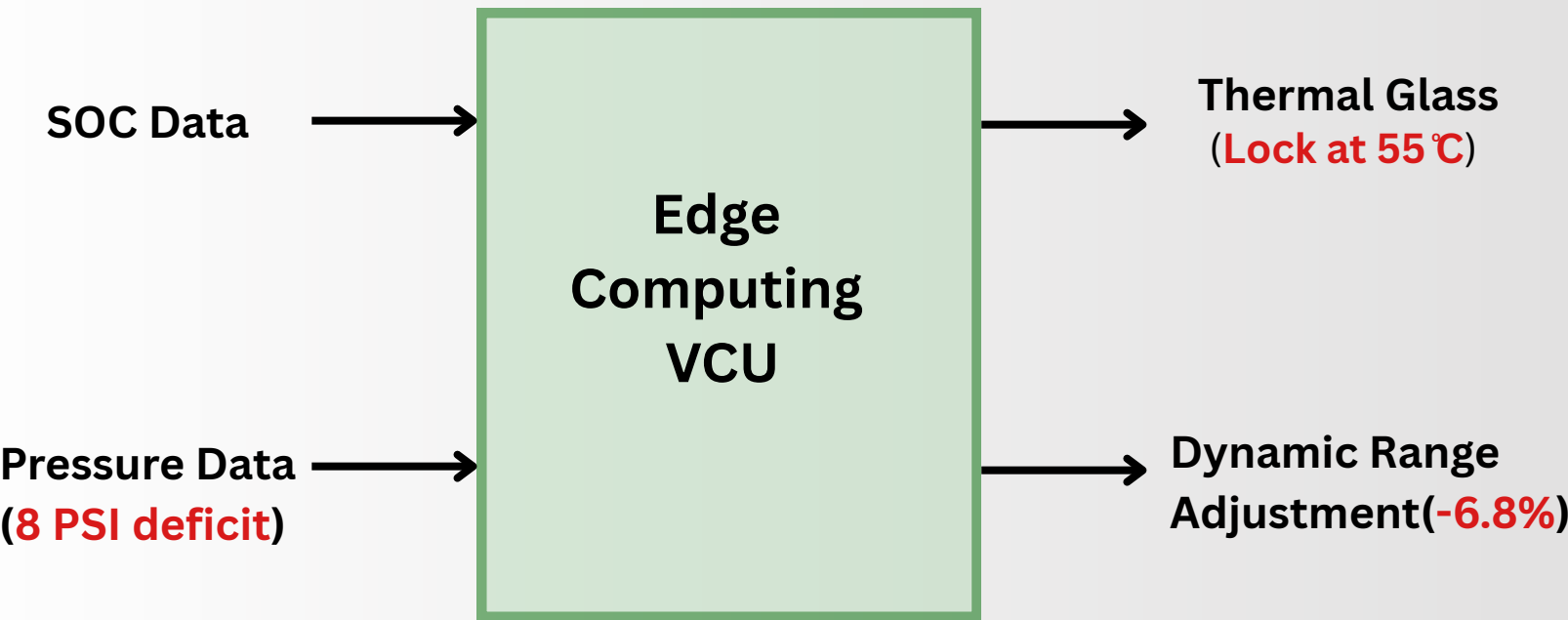


VRIO Strategic Analysis Framework for Hardware

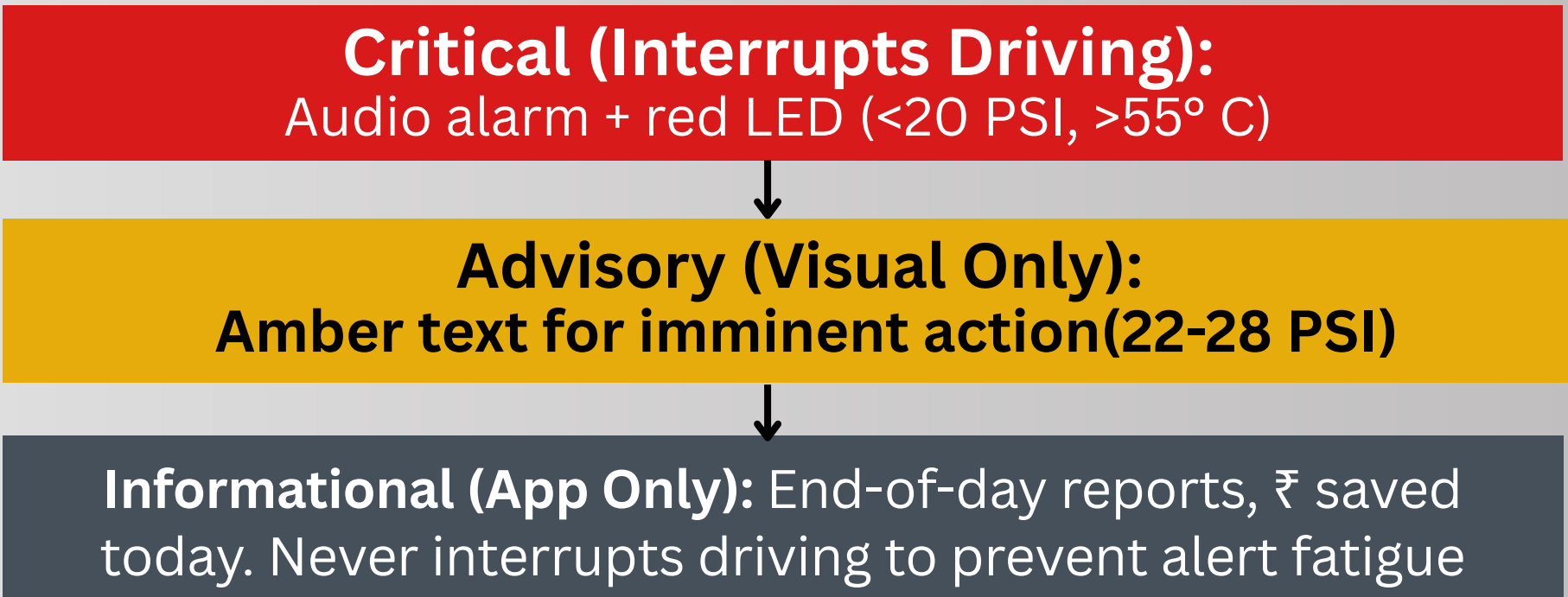
Value	Rarity	Imitability	Organization
Yes. Solves exact TCO leaks.	Yes. First natively integrated AC Fast + BLE TPMS on a 3W platform.	High Moat. Cloud QR rotation protocol and OCL algorithms cannot be copied by retrofits.	Yes. Bajaj's 5,000+ national network pre-positioned to deploy.

Intelligence & Fleet Platform

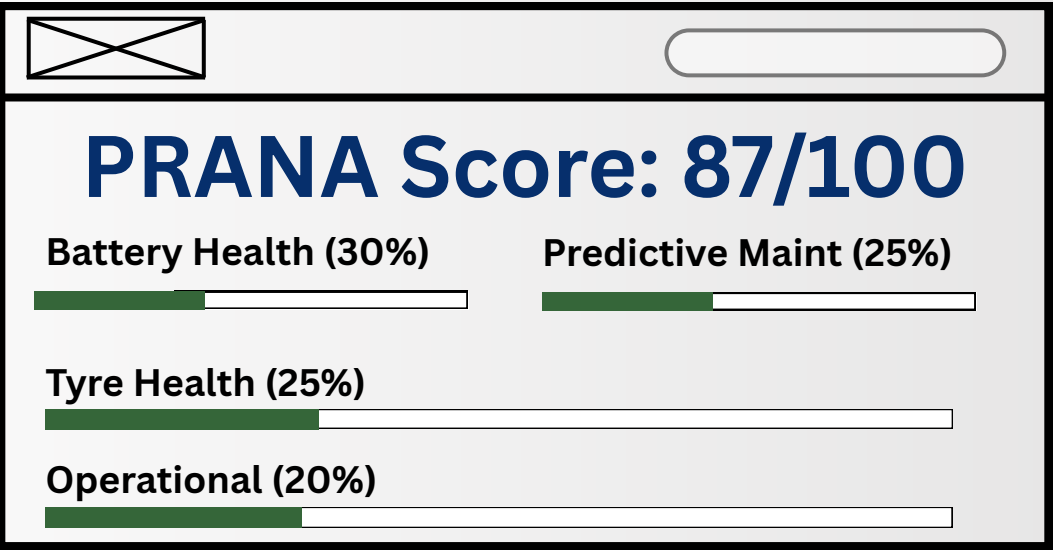
Sub-System 03: Vehicle Intelligence Layer (VIL)



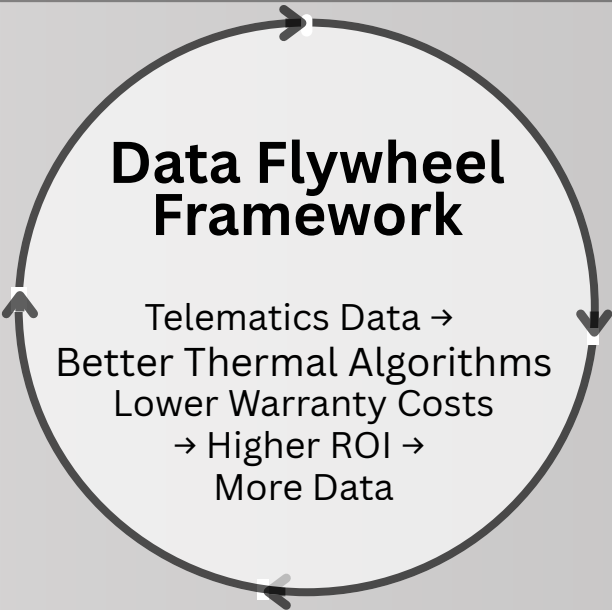
Operator Companion (OC) Three-Tier Logic



Fleet Intelligence Portal (FIP)



Smart Depot Management: Automatically staggers charging to avoid peak grid demand, saving **₹1,200-2800/vehicle/year**.



PQCDSM Trade-off Analysis

Cost (C) - 30%	Delivery/Uptime (D) - 25%	Safety (S) - 20%	Productivity (P) - 12%	Morale (M) - 8%	Quality (Q) - 5%
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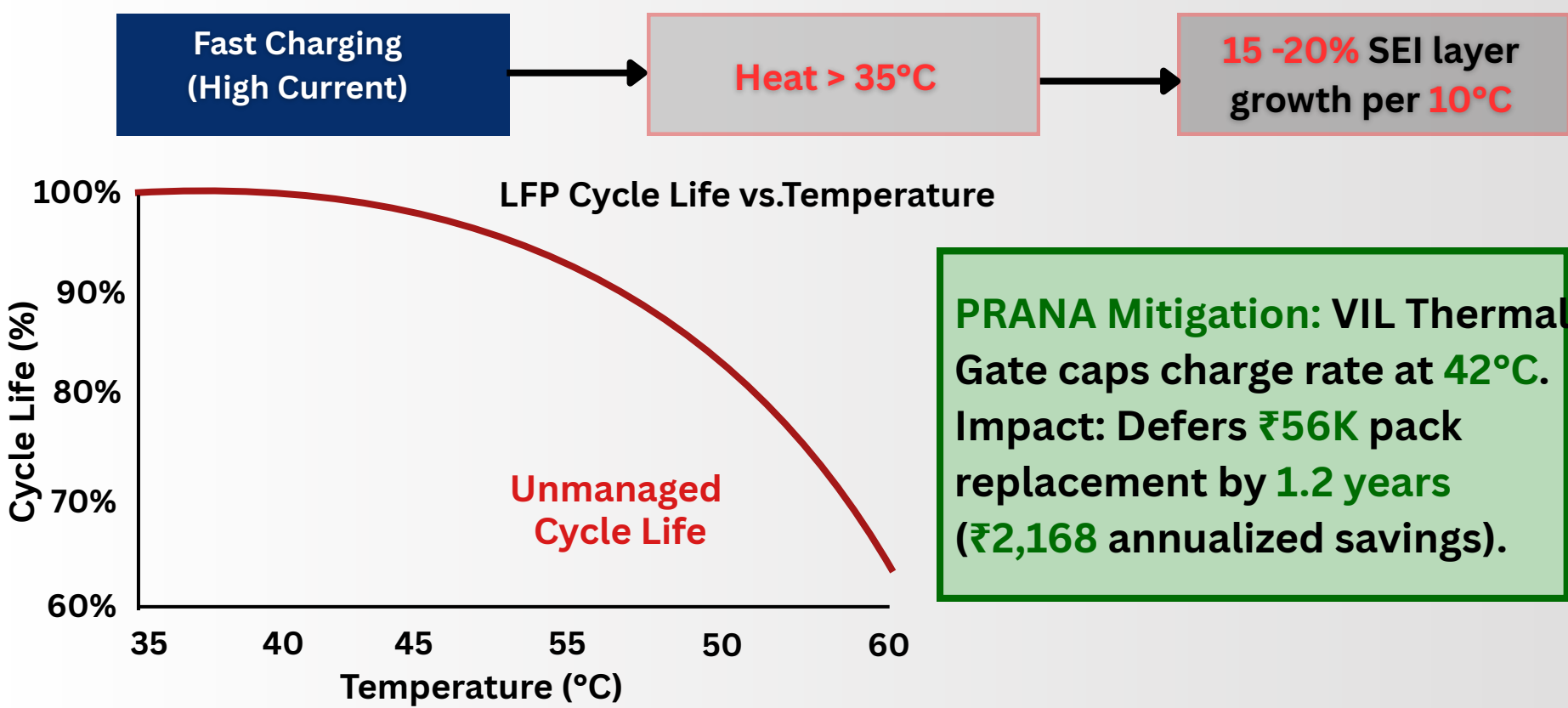
Charging Architecture PQCDSM Table					
	AC 3.3kW	DC Fast 15-25kW	Battery Swap	Total	Highlights
P	4.70	5.00	1.25	5.00	
Q	4.60	4.60	2.00	4.65	
C	5.00	3.50	2.00	3.00	
D	2.00	4.00	4.00	1.75	
S	1.70	2.50	3.00	2.00	
M	1.20	1.20	3.00	2.15	
Q	0.00	0.00	0.00	2.00	missing Tier 2
Total	5.00	4.65	4.67	2.00	missing tier 2 infra
Conclusion: AC 3.3kW mathematically outperforms due to universal and low thermal stress.					

TPMS Architecture PQCDSM Table					
	Direct BLE	Indirect	Hybrid	Total	Highlights
P	4.00	2.00	4.00	5.00	
Q	4.00	2.00	4.00	4.73	
C	3.00	1.50	4.00	2.50	adds extreme cost
D	1.70	1.00	3.75	2.50	
S	1.60	2.00	3.00	2.00	
M	1.20	1.60	2.00	1.75	
Q	0.00	0.00	0.00	2.00	non-existent ABS
Total	5.00	4.50	4.73	2.00	non-existent ABS
Conclusion: Direct BLE is the only technically viable solution for non-ABS platforms.					

Strategic Result : Tiering scales technology across all demographics without forcing entry-level buyers into premium software

Battery Physics & Validation

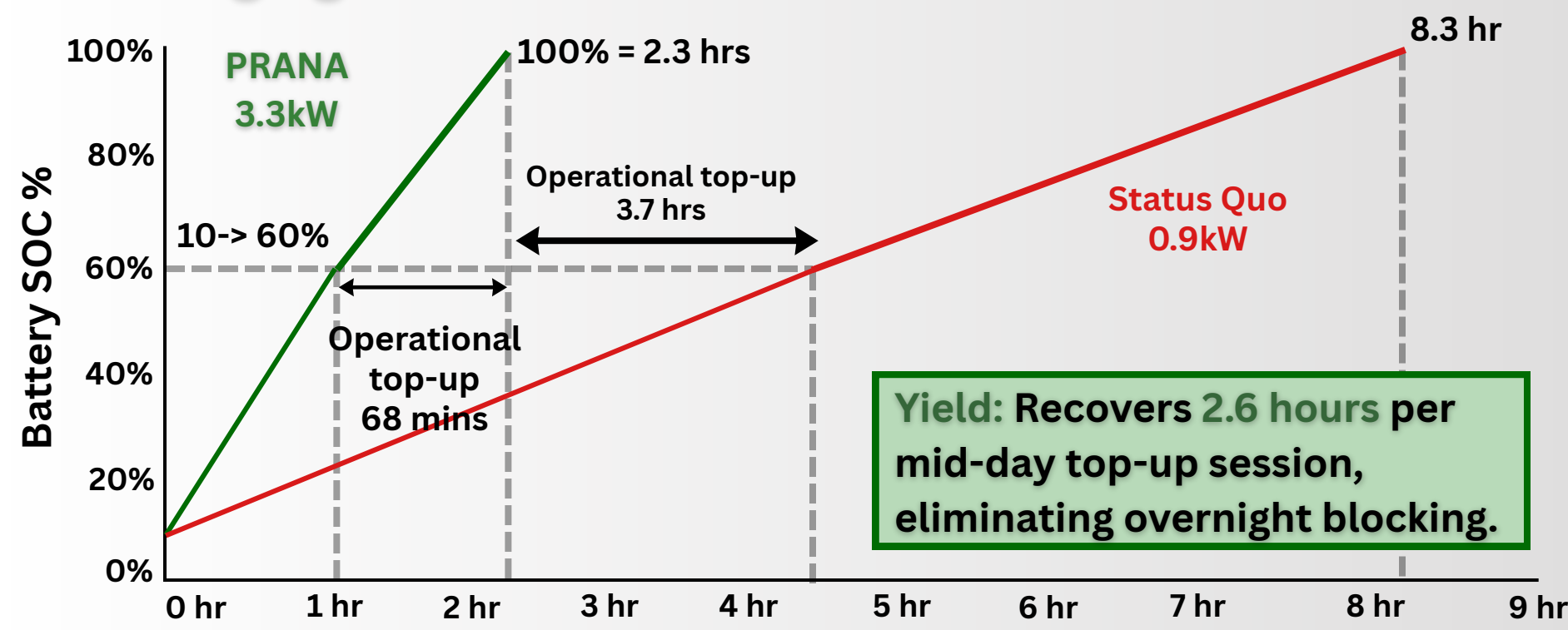
LFP Battery Degradation Physics



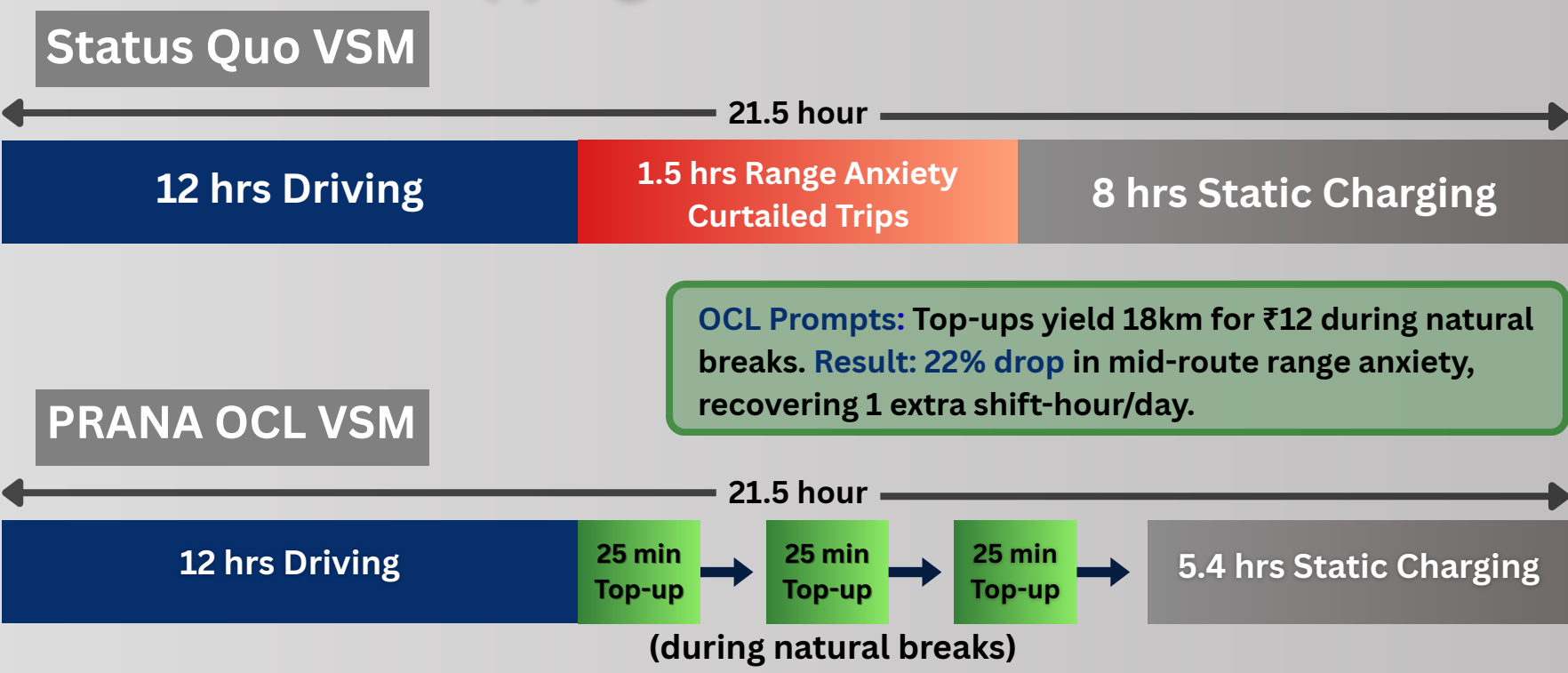
FMEA(Failure Mode & Effect Analysis)

Risk	Severity	PRANA Built-In Mitigation
Unmanaged Thermal Runaway	Critical	Hard VIL system lock at 55°C independent of cloud.
Post-Rotation Mis-mapping	High	App-guided QR Cloud reset enforcing >99.5% accuracy vs 34% manual failure.
Alert Fatigue	Medium	Strict 3-tier discipline; informational alerts suppressed during motion.

Charging Time vs. Downtime Math

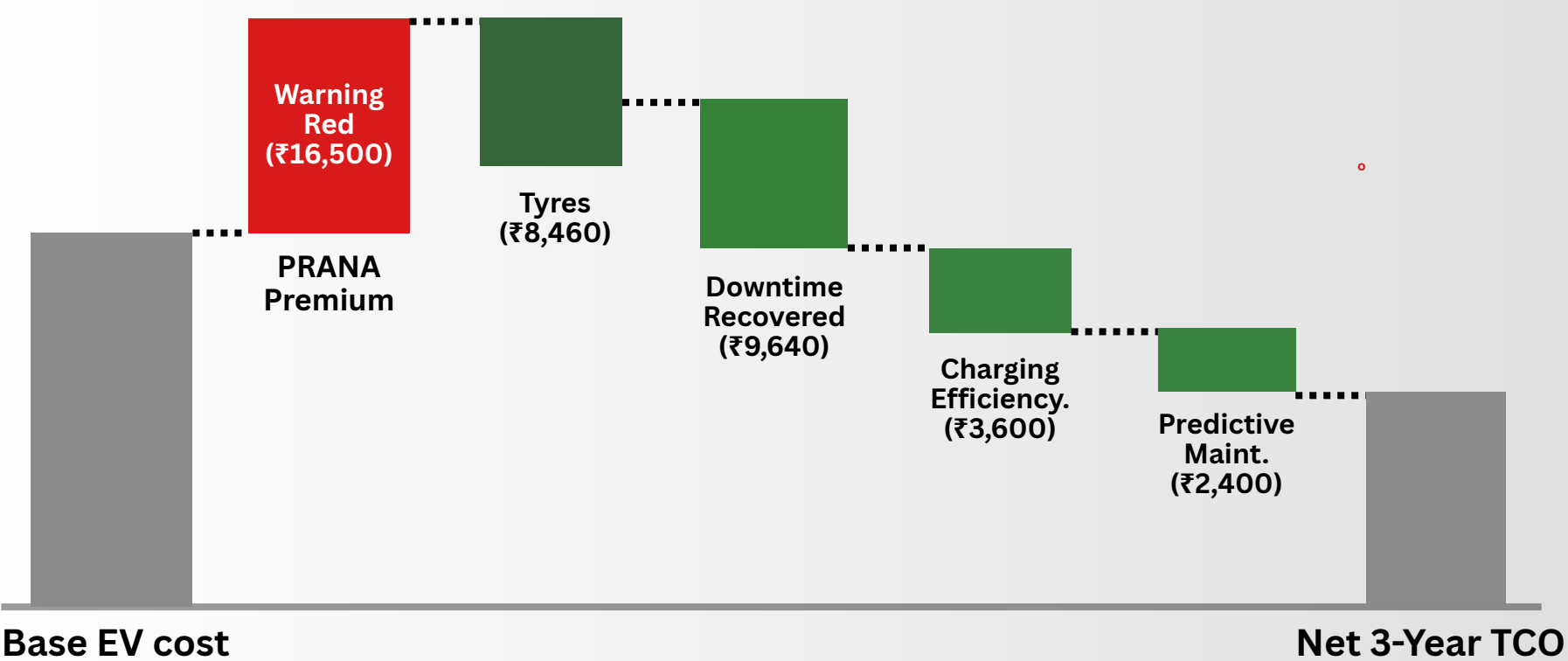


Value Stream Mapping (VSM)



TCO & ROI Modelling

TCO Waterfall Framework



Operator Savings Breakdown

Base Savings (₹24.2K/yr)

Blowout prevention (₹7.2K), range recovery (₹3.6K), downtime (₹5.7K).

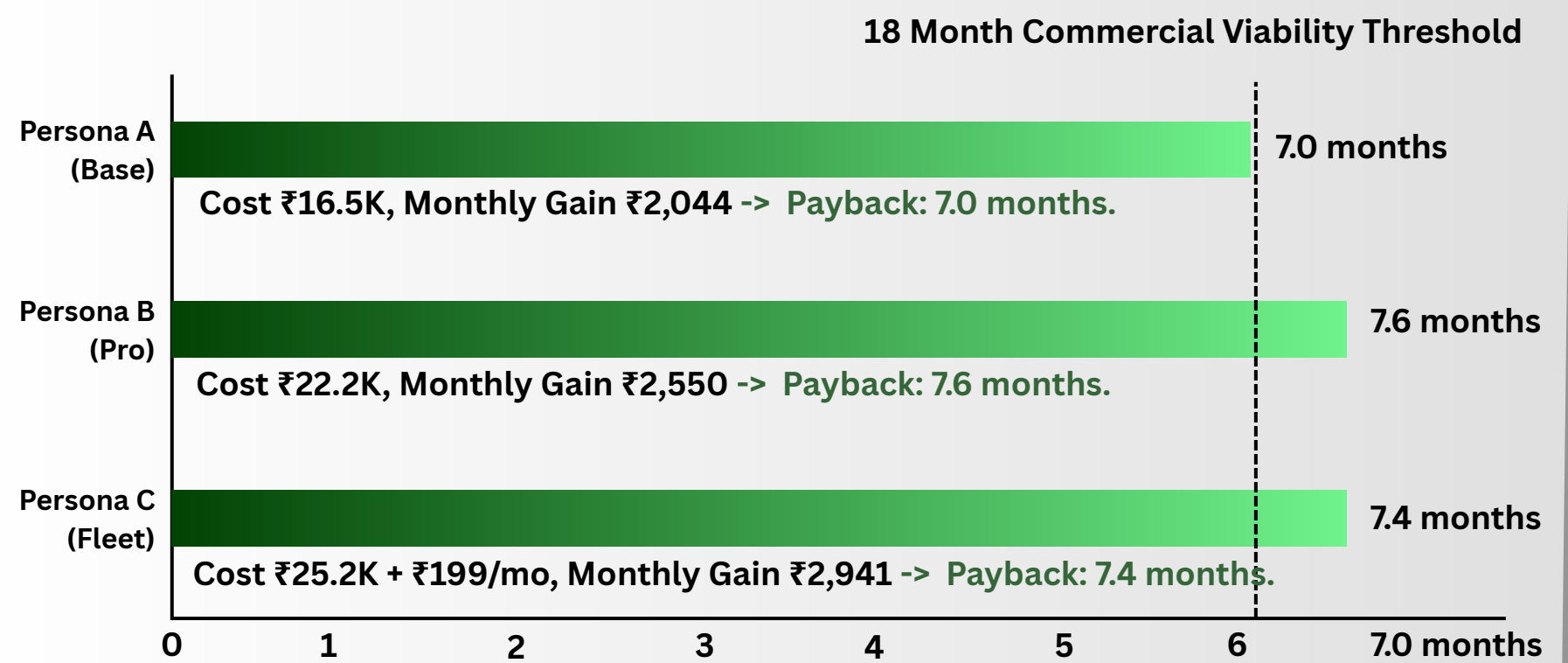
Pro Savings (₹30.8K/yr)

Adds smartphone OC analytics & route efficiency.

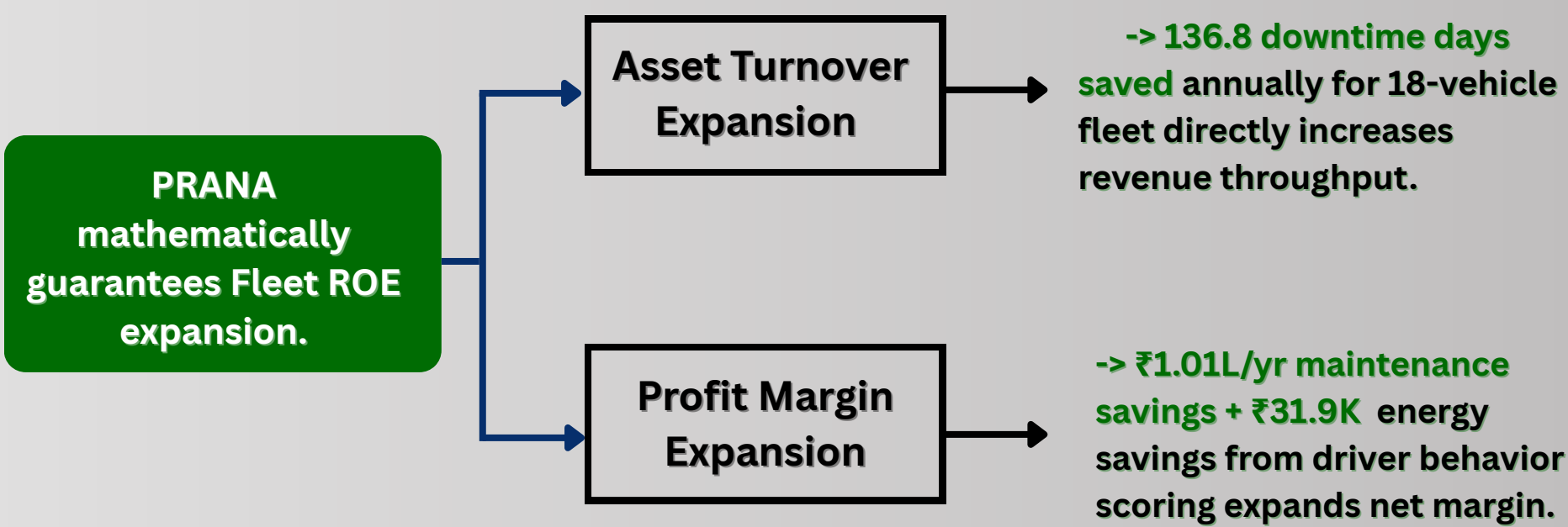
Fleet Savings (₹37.4K/yr)

Adds staggered charging (-₹1.2K peak grid charges) + behavior scoring (-₹31.9K fleet-wide energy savings).

ROI & Payback Period Modelling

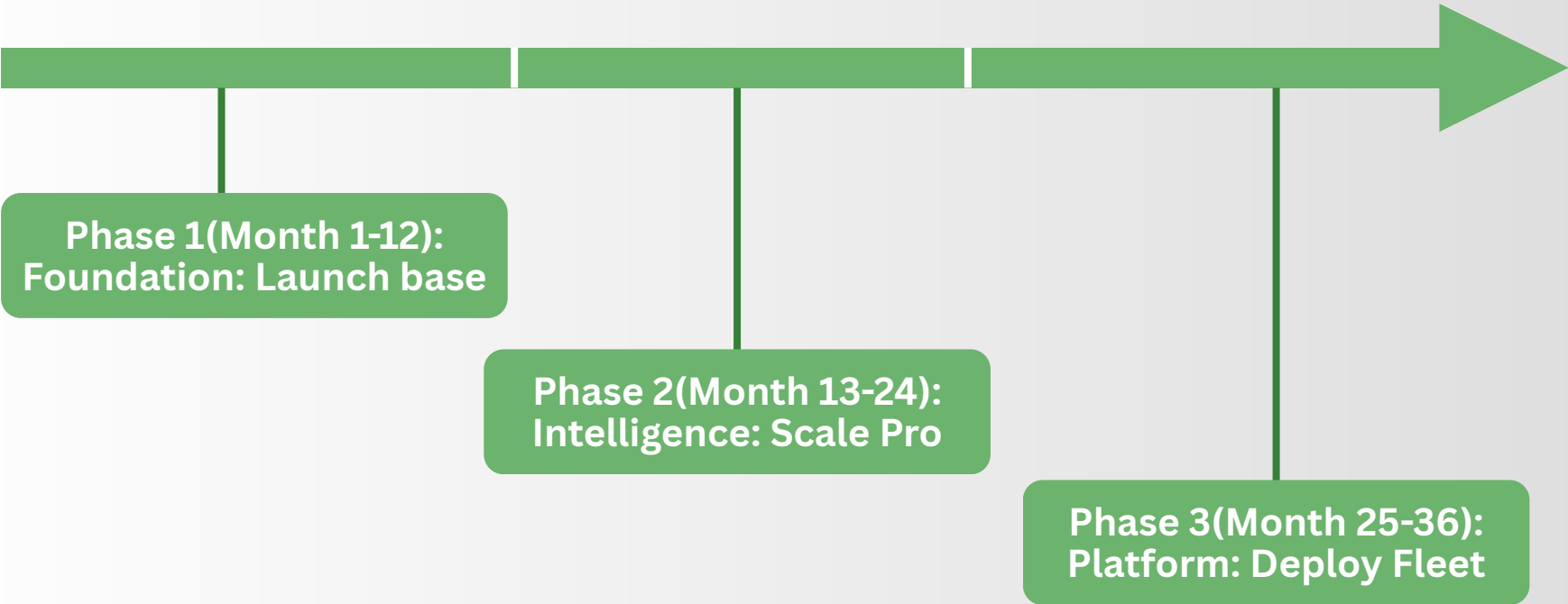


Dupont Analysis/ROI Tree Framework

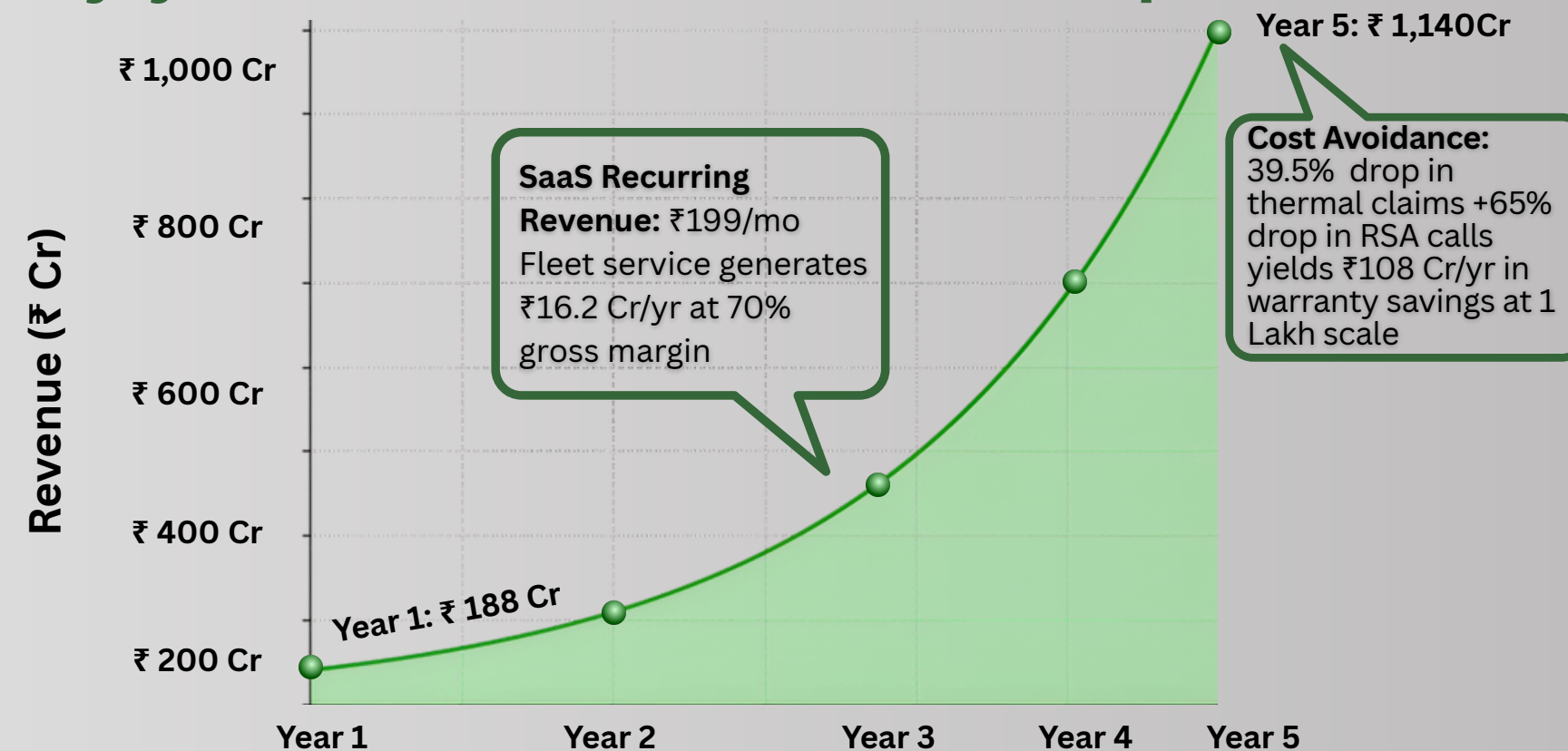


Implementation & Business Impact

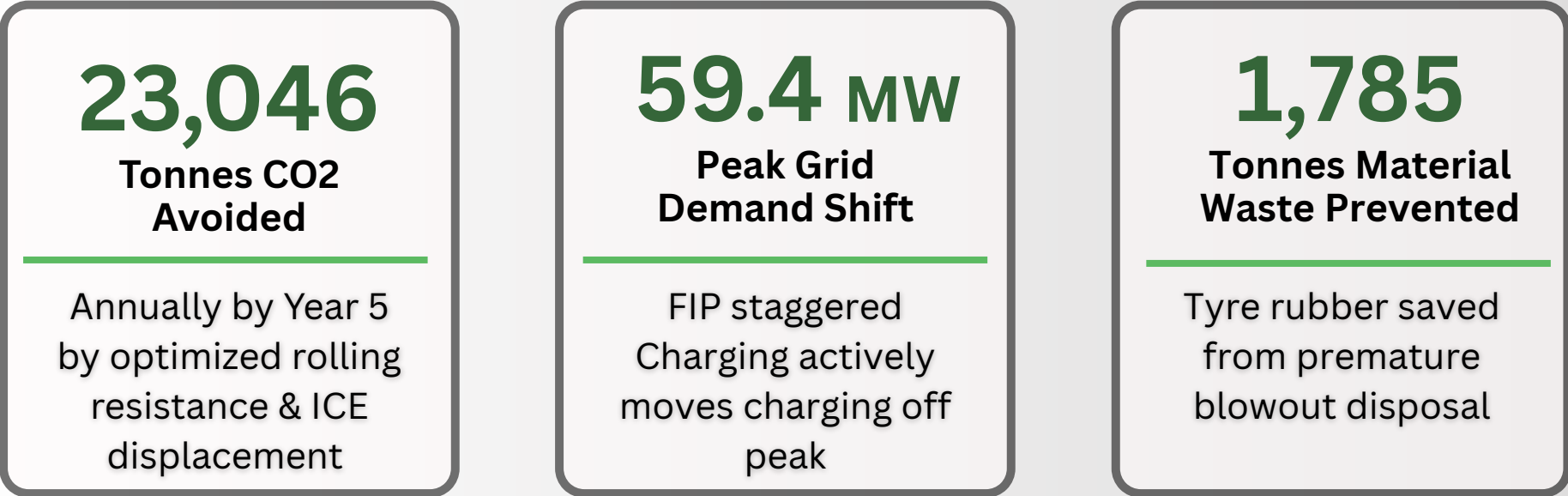
Implementation Gantt



Bajaj Business and Financial Impact



ESG METRICS



Balanced Scorecard Framework

